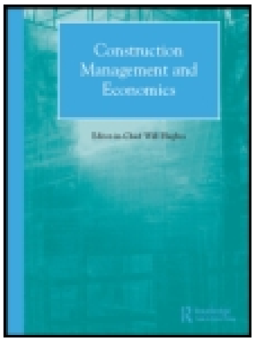

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EDITORIAL

Occupational health and safety in the construction industry

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The expectation that workers be able to work productively without suffering harm as a result of wealth-generating activities is the sign of a mature, responsible and equitable industry. The International Labour Organization (ILO) estimated that at least 60 000 fatal accidents occur each year on construction sites around the world, representing one fatal accident every 10 minutes. Construction accounts for one in every six fatal accidents recorded at work annually (International Labour Organization, 2005). Further, the ILO estimates that the construction sector in industrialized countries employs between 6% and 10% of the workforce but accounts for between 25% and 40% of work-related deaths. Equally significant is the impact of work on the health of construction workers. The ILO estimates that 30% of construction workers in some countries suffer from back pain or other musculoskeletal disorders. These are shocking statistics which speak to the need for a determined and coordinated effort to develop, implement and evaluate new approaches to address this intransigent problem.

The motivation for this special issue was a belief that policy makers, industry employers, union groups and researchers need to pay greater attention to integration and interface issues in the management of occupational health and safety (OHS) in the construction industry. The special issue was informed by the growing recognition that addressing OHS within a single enterprise is no longer sufficient to bring about sustained and significant improvement in OHS in the construction sector.

Construction projects are technologically and organizationally complex. Conventional OHS risk management methods assume that work can be 'decomposed' into its constituent parts. Each of these parts is then subject to a risk assessment and appropriate control measures are selected. The problem with this approach is that decomposition of complex systems is of limited value when system elements are in constant dynamic interaction with one another (Cooke-Davis *et al.*, 2007). It is practically impossible to predict whether such interaction will create new OHS hazards at the interface between organizations, technologies, components or activities. If undetected, these hazards are uncontrolled.

In construction, there is a need to manage the interests and influences of multiple project contributors and stakeholders who, either consciously or inadvertently, exert an influence on OHS.

Coordinating the activities of different contributors to the design and construction of a facility can be challenging because each participant will be influenced by their interactions with other project participants, while also pursuing their own individual or organizational business interests. There is a need to ensure compatibility between the components that make up a facility and to ensure that the construction and installation methods implicit in the design of the parts does not produce unanticipated hazards or unacceptable risk at their interfaces. The activities of different work crews and trades also need careful management to ensure that work processes are sequenced to minimize the possibility of adverse impact. The physical worksite environment itself creates coordination challenges, as workers, materials and equipment are constantly moving in the site environment and the landscape changes daily. The backdrop to these challenges is the labour-intensive nature of construction work and the heavy reliance on subcontracting. This has created a situation in which employment for construction workers is insecure and opportunities to engage in collective bargaining and

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OHS training are limited. Research indicates that those at the bottom of the contractual chain bear the most significant consequences when things go wrong (Swuste *et al.*, 2012).

This special issue presents a collection of thought-provoking research papers, each of which addresses the challenge of integrating OHS into construction project management. Several of the papers speak directly to the challenge of managing OHS at the interfaces between project participants and/or technologies. A number of the papers provide insight into how OHS is spoken about and understood at construction workplaces and several papers describe new and valuable ways to understand workplace cultures and the critical role of OHS communication and meaningful consultation between managers and workers. It is hoped that this special issue will stimulate further research and debate which will eventually inform a fuller integration of OHS into all aspects of construction project management. The special issue highlights the opportunities to engage different theoretical perspectives and methodological approaches that will be useful in supporting this integration. However, all of the papers reveal how the integration of OHS into construction project management requires a more sophisticated and nuanced understanding of the way in which the construction industry's inherently complex products and processes impact on OHS.

Supply chains and networks

Quinlan (2011) argues that OHS problems arise in supply chains and production networks as a result of three factors, namely: (1) economic and reward pressures that become successively greater towards the bottom of supply chains; (2) disorganization due to the engagement of many different (often small) businesses; and (3) the use of workers, whose employment is often precarious, working within complex and fragmented production arrangements.

Within the construction industry, the supply chain is more fragmented than in most industries. Projects involve many different professional and technical inputs and contributors are arranged in complex and hierarchical commercial configurations. Traditional procurement methods create structural challenges for the consideration of construction OHS issues during the preconstruction planning and design stages. Yet, while more integrated procurement approaches provide greater opportunity to address construction OHS earlier in the project life cycle, there are no guarantees that this will happen (Atkinson and Westall, 2010).

One potentially useful approach to understanding the importance of integrating mechanisms in supporting OHS improvements in the construction industry is offered by Lawrence and Lorsch (1967). They describe how organizations differentiate into distinct sub-systems as they grow. This is a necessary state of affairs as people and groups develop specialized skills and roles. However, Lawrence and Lorsch also point out that functional differentiation can entrench distinct (even conflicting) orientations towards organizational goals. They observe how people within different functional groups develop significantly different work processes and priorities. Lawrence and Lorsch explain that, if an organizational system is to remain viable, there must be certain mechanisms for integrating the different functional groups. These integrating mechanisms are needed to ensure that organizational members develop some degree of unity of effort in relation to organizational goals (Lawrence and Lorsch, 1967).

Writing about behaviour within construction projects organizations, Walker (2007) contends that members of distinct functional groups develop specific responses to a whole range of project objectives. These responses are sometimes at odds with one another. Therefore it is possible that some functional groups may be strongly focused on OHS, while other functional groups are less so. Tensions between safety and production efficiency have, for example, been noted. The phenomenon of group differences in safety focus has been observed in construction projects. For example, Gherardi *et al.* (1998) describe functional and professional differences in construction industry participants' safety responses.

Differentiation in construction projects takes several forms. Project participants are divided along organizational, technological and cultural grounds.

Organizational differentiation exists as a result of commercial and contractual arrangements in which many different contributors are engaged at different times, under different contracts, to design and deliver different aspects of a project. The commercial arrangements under which contributors are engaged vary considerably from project to project and can have a significant impact on the opportunities presented to integrate OHS into decision-making, especially in preconstruction project stages.

Technological differentiation exists because product complexity and technological development have created the situation in which many different specialist designers, building component manufacturers and installers are engaged in the design and installation of particular building components. While OHS considerations may be well integrated into the design and installation of a specific component, hazards may arise as a result of the interaction between components and their installation.

Finally, cultural differentiation exists as a result of variation in the OHS orientation of different industry contributors and stakeholders. Research has demonstrated that cultural uniformity in relation to OHS does not exist in organizations. This is particularly likely to be the case when, as in construction, productive work is performed in locations removed from corporate offices. Further, training and socialization processes are likely to give rise to professional differences in values, beliefs, attitudes, customs and behaviours related to OHS.

In the absence of strong integrating mechanisms, differentiation can produce deep fissures, conflicting interests, communication failures and a lack of clarity concerning responsibility for OHS. Thus, the construction industry's fragmented supply chain militates against the effective management of OHS in construction projects (Swuste *et al.*, 2012).

The ages of safety: do we need to go back to the future?

Hale and Hovden (1998) describe three distinct 'ages' in the evolution of OHS. The first age of safety responded to fundamental changes in agricultural, industrial and manufacturing processes that began with the industrial revolution. According to Hale and Hovden (1998), the first age of safety was principally focused on the development of engineering and technological solutions to a new set of workplace hazards. However, limitations inherent in a purely technological approach came to be recognized and attention shifted to the interface between people and technology. Thus, the second age of safety saw a greater focus on human factors. However, towards the end of the twentieth century, the focus again shifted. Research had begun to reveal the critical role of management and organizational factors in shaping OHS outcomes. Consequently, Hale and Hovden (1998) describe how, in the third age of safety, the focus has been on the implementation of OHS management systems.

While these ages of safety are sometimes understood to be mutually exclusive, Glendon *et al.* (2006) argue that the identification of distinct periods in the evolution of OHS does not imply that previous periods were superseded. Rather, they assert that each period builds upon its predecessor, creating the possibility of multiple and more complex OHS strategies. They define this as the 'fourth age' of OHS in which the potential benefit arising from the integration of multiple approaches is recognized. This is an important point because each period in the development of OHS provides potentially useful tools and approaches. For example, the emphasis on the man–machine interface remains critical in informing safety in design, while the implementation of OHS management systems ensures a consistent and programmatic approach to planning, resourcing, monitoring and improving OHS. However, technological interventions perhaps provide the most significant potential for eliminating OHS hazards or reducing risks at source. Indeed, there is widespread acceptance that technological methods available for controlling OHS risk (i.e. those that eliminate a hazard or provide for physical separation between the hazard and people who may come into contact with it) are more effective than controls that rely for their effectiveness on workers' behaviour, such as administrative measures or personal protective equipment. This thinking is often represented as a hierarchical arrangement of risk control options, reflecting their relative levels of effectiveness.

With this in mind, it is useful to consider the circumstances in which the opportunity to develop robust technological solutions to construction is influenced by organizational and cultural factors.

The fourth age of safety (the so-called 'integration age') presents exciting opportunities for deploying advanced technologies to improve OHS in construction. However, new technologies need to be developed with a good understanding of the social arrangements of production within which they are to be used. Recent decades have seen a significant growth in research and development of innovative technology-based OHS tools (see, for example, the review of advanced technology and OHS provided by Zhou, Irizarry and Li in this issue). Also in this issue, several papers describe the application of specific forms of advanced technology to improve OHS. Melzner, Zhang, Teizer and Bargstädt describe the application of BIM software to identify and resolve safety hazards during the planning of construction work. Also, in two separate papers, Li, Chan and Skitmore and Marks and Teizer describe the deployment of real time proximity detection systems to reduce the risk of 'struck-by' accidents in the dynamic construction worksite environment.

Preliminary evaluations of the usefulness of these technologies suggest they have the potential to improve OHS performance. However, as Harty (2008) eloquently demonstrates, prevailing social structures and organizational arrangements are key factors in determining whether new technologies are adopted into practice. Thus, the effective deployment of technological solutions to the industry's OHS problems may be greatly enhanced by the adoption of a socio-organizational-technical systems theoretical perspective to assess and optimize the extent to which there exists a good 'fit' between people, the tasks they perform and the new technology.

Safety systems: barriers to effectiveness

Many organizations, including large construction organizations, have responded to OHS challenges by implementing occupational health and safety management systems (OHSMSs), though the effectiveness of these systems has been questioned.

Gallagher *et al.* (2001) describe how, to be effective, OHSMSs must be integrated into broader organizational management systems. Such integration is addressed by Forman in this special issue. Exploring the nexus between OHS management and lean construction, she reveals how historical and contextual factors influence the linking of lean construction and OHS. In particular, the position and influence of key organizational actors is critical to the integration of these management approaches. Gallagher *et al.* (2001) argue that the level of integration of OHS management activities into general business management is often insufficient to ensure OHS is adequately resourced and managed. They observe that OHS is too often regarded as the responsibility of a specialist functional department and the management of OHS is divorced from operational or strategic decision-making within organizations.

Frick and Wren (2000) also identify a tendency for OHSMSs to be overly complex and cumbersome. In some cases, the documentation of the OHS management process can actually detract from less formal but more effective workplace strategies for dealing with OHS.

In some instances the production of documentation required by OHSMSs becomes an 'end' in itself, bearing little relation to how work is actually carried out. In Australia, for example, there is mounting criticism of Safe Work Method Statements (SWMSs), which are mandatory for high risk construction work. The development of impenetrable and unnecessarily lengthy SWMSs (some are reportedly 70 pages or more!) is unhelpful in an industry in which literacy levels are lower than average. Anecdotal evidence suggests that it is common for workers to commence work without even seeing the SWMS.

Not only do workers have legal and moral rights to be informed about the OHS aspects of their job, but frequent and regular OHS communication is important in maintaining focus and vigilance. Alsamadani, Hallowell and Javernick-Will address the critical issue of OHS communication in this special issue. The authors reveal that work crews with better than average OHS performance received at least weekly formal OHS communication from management and also engage in weekly informal OHS communication within their work groups. Most tellingly, the crews with the highest OHS performance engaged in all forms of communication, i.e. communication was not always in written form, but also included toolbox talks and informal conversations.

Systems or cultures

Hopkins (2005) argues that a system alone does not ensure OHS performance because systems need to be embedded in the right type of organizational culture. Glendon and Stanton (2000) describe two types of safety culture, i.e., a functionalist, 'top-down' culture and an interpretive, 'bottom-up' culture. The former approach views culture as something to be assessed and modified by management. In a functionalist safety culture, uniformity is pursued through the hierarchical exercise of top-down control. In contrast, an interpretive approach recognizes culture to be an emergent and complex phenomenon, which is not owned or controlled by a single group, but is created through interactions between all members of an organization (see, for example, Richter and Koch, 2004). Koch adopts an interpretive approach to safety culture in his paper in this special issue. He builds on his previous work to reveal how the collective identity, beliefs and behaviours of work crews in the construction industry are overlaid and influenced by complex constellations of national culture. Thus Koch's paper demonstrates that safety cultures are multi-faceted, complex and not confined within organizational boundaries.

The dangers associated with failing to fully appreciate local manifestations of safety culture are clearly illustrated in the paper by Tutt, Pink, Dainty and Gibb in this special issue. It is often assumed that migrant workers are vulnerable in terms of OHS because they are unable to access and understand important OHS information in the English language (see, for example, Trajovski and Loosemore, 2006; Bust *et al.*, 2008). In their paper Tutt *et al.* describe the emergent culture of a construction work crew, comprising members from several Eastern European countries. The authors describe how the crew members developed their own unique 'language' for communicating OHS information. The crew was repeatedly recognized by the principal contractor at the worksite as demonstrating the highest levels of safety, despite not using English to communicate. This case reveals that attempts to exercise top-down cultural control could potentially disrupt existing and highly effective work practices. Rather than strive for a uniform culture driven by a dominant group (i.e. the English-speaking site managers) the work reveals the value of acknowledging and reinforcing locally good practice.

Hopkins (2005) describes 'mindful' leadership as a key component of safety culture. According to Hopkins (2005) the state of mindfulness involves maintaining a constant sense of unease about the state of health and safety in an organization. Mindful leaders also make frequent visits to worksites and actively engage with and listen to what workers have to say about health and safety. In this issue, Sherratt, Farrell and Noble reveal an underlying discourse of enforcement, rather than engagement, in the way in which OHS is written and spoken about at construction sites. Despite a growing appreciation that workers are an untapped source of good ideas for innovative and effective OHS solutions, Sherratt and her colleagues demonstrate that management structures remain hierarchical and focused on top-down behavioural control.

There is significant potential to unlock OHS knowledge by engaging in meaningful consultation with workers. The collective knowledge and experience of the workforce can be invaluable in designing effective and practical control measures for OHS risk. Further, as Walters and Nichols (2009) note, there are legal, moral and ethical imperatives for ensuring that workers have the right to be represented in health and safety matters. Legal rights to representation are a key feature of Australian OHS legislation (Johnstone, 2009), but Walters and Frick (2000) describe how workers' participation in OHS management processes can take many forms. It is imperative to consider who participates, how much and why, as these variables determine how effective the participation will be in relation to reducing OHS risks. In this issue, Ayers, Culvenor, Sillitoe and Else describe research investigating the quality and impact of workers' participation in OHS in the Australian state of Victoria. They report low levels of organizational maturity in terms of engaging workers in consultation in relation to OHS. The authors report that consultation in the Victorian commercial construction industry is limited to the resolution of day-to-day operational issues. The potential benefits to be gained from engaging workers in more strategic OHS decision-making are significant, but are currently not being realized.

Integrating health and safety into planning and design

In construction OHS research and policy development there has been a growing recognition that some responsibility for health and safety should rest with players upstream in the industry's supply chain. Notably, there is now a widespread acceptance that decisions made in the planning and design stages of construction projects can have health and safety implications in the construction stage (see also the paper by Behm and Schneller in this issue).

The hazard prevention through design (PtD) movement places an emphasis on designing OHS hazards out of the construction industry's products and processes altogether. However, ensuring that OHS knowledge is available and used to inform decisions made in the planning and design stages of projects remains a challenge for the construction industry. A commonly cited problem is the level of construction OHS knowledge possessed by upstream decision-makers. There is a need to ensure that preconstruction decision-makers are better informed about OHS to enable them to understand more perceptively the risk implications of their decisions. OHS is an area where learning from one's mistakes is the least desirable method of learning and, thus, integrating OHS knowledge into the education and qualifications of upstream construction industry professionals has been recommended (see, for example, Toole, 2005). An alternative approach is to capture OHS knowledge and make this knowledge available to decision-makers in the form of a decision support system. In this issue, Esmaeili and Hallowell describe a decision support system that profiles OHS risk in typical highway construction activities. The tool enables those engaged in preconstruction planning and scheduling decisions to understand the OHS implications of the proposed construction programme prior to commencement of work.

While the viability of designing for construction workers' health and safety has been demonstrated in certain circumstances (see, for example, Gambatese *et al.*, 2005), practical difficulties in the implementation of PtD in the construction industry remain. Design work in the construction industry is extremely complex and attempts to attribute responsibility for health and safety to the occupant of an abstract sociotechnical role, i.e. 'the designer' is not as easy as the legislation and policy documents would suggest. Lingard *et al.* (2012) use case studies to illustrate how the multiple inter-related tasks undertaken by design specialists within a complicated 'web' of inter-organizational relationships are not well suited to a standardized risk management approach. Indeed, attributing responsibility for design decisions that impact on OHS is extremely difficult because many aspects of design are not performed by principal design consultants and stakeholders external to the project team can also play a significant role in influencing design decisions. Wright *et al.* (2003) note that many PtD solutions are driven by building systems manufacturers, and not principal design consultants. This reinforces the argument that an integrated supply chain approach to the management of OHS is required in the construction industry.

The dynamic nature of construction design work also presents challenges to the management of OHS. In this issue, Larsen and Whyte present evidence to support this contingent view of construction design. They show how iterative design changes can impact on construction workers' OHS.

PtD policy in the construction industry been challenged on the basis of the lack of clarity concerning what is actually being designed. For example, Driscoll *et al.* (2008) reviewed the findings of coronial investigations in Australia to identify the extent to which design was a causal factor. They report that 44% of the construction industry deaths examined were design-related. However, a closer inspection of the accident circumstances described in the report reveals that the majority of the deaths were related to the design of plant or electrical equipment being used at the time or to design of the process of work (including temporary works). Indeed, only one of the fatal accidents analysed by Driscoll *et al.* (2008) was clearly related to the design of the permanent structure. This death involved a maintenance worker falling through a fragile skylight while working on the roof of a building. It is also apparent that many of the commonly cited PtD solutions in the construction industry actually involve a redesign of the construction process, rather than the permanent building or structure to be constructed. Atkinson and Westall (2010) note, with some disappointment, that many design modifications implemented to improve OHS construction projects represent fairly modest solutions. They cite examples of fixing rails or anchor points for fall arrest devices, which do not eliminate an inherently dangerous activity, i.e. working at height. Modification to the design of a process to reduce OHS risk is certainly worthwhile but this should be distinguished from consideration of safety in the design of the permanent facility to be constructed. This distinction has important implications for the implementation of PtD because reducing OHS risk through product and process design would require intervention at different points in the project life cycle and the involvement of different participants. It may also be the case that the best OHS risk control measures are achieved when the industry's product and process design are considered simultaneously.

The construction industry is characterized by high levels of technological, cultural and organizational differentiation which create substantial challenges for the management of OHS. OHS hazards can arise as a result of interface issues, poor coordination, separation between planning, design and construction functions, variation in the OHS orientation of different technical and professional groups and dissonance between public statements contained in OHSMS documentation and worksite practice. Mainstream project management practices which decompose a complex process or building into component activities or elements militate against the development of a holistic approach to the management of OHS risk. The construction industry's OHS performance appears to be resistant to change, suggesting the need for a fresh approach. Given the industry's problems, further OHS improvements may well require the conscious development of effective integrating mechanisms.

In this issue

Each of the 13 papers in this special issue addresses integrating mechanisms to ensure that differentiation in the construction industry does not negatively impact on construction workers' health and safety. The papers showcase the diversity and depth of international construction OHS research and offer many insights into ways in which improvements in the industry's OHS performance may be realized.

Tutt, Pink, Dainty and Gibb examined the communication of health and safety information between migrant workers and English-speaking site management in the UK construction industry. The authors encountered the apparently contradictory situation of a trade team on a large project topping the site safety tables, while not speaking English in much of their everyday work. This invited the authors to probe the assumptions (and factors) of what makes safe practice and effective communication on site and how this is achieved. They study how a team of curtain wall installers had evolved its 'own GlazaBuild language': a conglomerate of communication methods, a mix of different languages, gestures, simple hand signals and phone links which together helped coordinate a series of complex tasks. The authors demonstrate how an ethnographic approach can enable an understanding of how communication channels are developed by migrant workers within the specificity of a gang, their strategies and successes in staying safe and the ways of knowing/knowledge this involves. Tutt *et al.* argue that attempts to develop a culture of safe working should seek to identify, embrace and complement the practices where local/tacit knowledge is enacted, rather than necessarily seek to displace them or to generate oversimplified generalizations.

Esmaeili and Hallowell consider the extent to which safety programme elements are integrated into decision-making during the programming and preconstruction phases of highway construction projects. To enhance preconstruction safety practices, Esmaeili and Hallowell quantified safety risks of highway construction and maintenance tasks, and developed and tested a decision support system to integrate safety risk data into the project schedules. Validation revealed that the database and decision support system have the potential to improve safety resource optimization and preconstruction safety management. Their practical contribution is to generate reliable safety risk profiles based on a risk database by considering temporal and spatial interactions among construction activities. By analysing the risk profiles, a project manager can use schedule float to distribute or concentrate risk during the life cycle of a project. In addition, risk profiles enable thoughtful safety planning and effective allocation of safety resources in a single project or multiple projects. For example, project managers can identify high risk periods and plan for extra precautionary measures (e.g. lane closure), develop customized injury prevention strategies or, at a minimum, inform workers of the tasks and interactions known to cause high risk periods.

Ayers, Culvenor, Sillitoe and Else explore the factors that constitute meaningful and effective consultation in the construction industry. They examined how senior site managers and worker OHS representatives of five construction contracting companies engaged in the commercial and industrial sector of the Victorian construction industry enacted legally mandated consultation processes and examined whether these consultation processes could be considered to be meaningful and effective. The companies were each allocated a level of organizational maturity, depending on how they managed various aspects of their business operations in terms of OHS. Senior managers and OHS representatives were chosen as participants in the research because they are generally acknowledged as the critical vectors in the sharing and transferring of knowledge and skill at the workplace. Utilizing an ethnographic methodology and implementing a combination of semi-structured interviews, non-participant observations and the study of documentation, the authors found that regardless of the level of organizational maturity, and no matter how individual participants applied particular moral and ethical principles, OHS consultation was limited to and focused primarily on day-to-day operational and execution aspects of the job. Strategic and/or longer term OHS issues were not part of the consultation forums. The authors created a model identifying the main issues that need to be addressed if meaningful and effective OHS consultation is to take place between senior managers and OHS representatives.

Alsamadani, Hallowell and Javernick-Will measure and model safety communication among small project crews on active projects in the Rocky Mountain region of the USA. Interviews were used to collect empirical data. Social network analysis (SNA) was employed to provide measures of knowledge exchange, including actor centrality, actor betweenness and network density. Network characteristics were compared between crews with a high and low relative safety performance. Additionally, network diagrams for high-performing crews were created and visually analysed for patterns. The results showed that crew managers in high-performing crews communicate safety issues through formal addresses with their personnel at least weekly, hold monthly training meetings, and communicate with their crew members informally on a regular basis. Among all SNA metrics, network density was the only significant measure that distinguished overall crew safety. Because previous literature has shown safety communication to be a key component of safety culture, the results will be useful as a leading indicator of safety performance. Future researchers can use the research methods described to structure further safety communication studies.

Behm and Schneller applied the Loughborough Construction Accident Causality (ConAC) model and propose it as a useful organizational learning tool for construction safety research and practice. They utilized the model to investigate incidents with a State Department of Transportation in the USA. Operational definitions were developed for each cause, factor and influence within the ConAC model which enhanced its reliability across each investigation. A detailed interview protocol was developed and is described so the study can be duplicated. The authors conducted semi-structured interviews with employees, witnesses, supervisors, managers and safety engineers to determine causes, factors and influences in 27 incidents. The ConAC model proved to be a useful tool to guide investigators who wish to extend their investigations beyond the identification of immediate causes (acts and conditions) and learn about the impact of upstream factors and influences (e.g. supervision, site constraints, safety culture, project management, etc.). Safety-related incidents are complex and multi-faceted. Construction organizations seeking to learn how to effectively reduce safety risks through incident investigations should view investigation as a research-based activity, using an appropriate theoretical model to identify upstream as well as immediate contributing factors. The authors demonstrate that the ConAC model can be a useful tool to support organizational learning in relation to safety.

Li, Chan and Skitmore integrate the use of real time positioning systems to improve safety in blind lifting and loading operations. Having carried out a case study and interviews in Hong Kong, the authors explored the risks associated with blind lifting and loading. They developed a safety support system integrating the use of positioning systems to obtain the precise position of workers and the crane's hook. The system presents real time information about the position of workers in relation to the crane hook in the loading and lifting zone. The authors applied the system in a case study construction site in Hong Kong. Interviews were conducted and site staff provided positive feedback regarding the use of the system. The authors suggest the system has the potential to prevent 'struck-by' accidents during blind lifting and loading operation of cranes. The use of positioning systems to monitor the location of workers or different objects represents an innovative use of advanced technology. The case study establishes the foundation for further development and application of positioning system technology for other safety management purposes.

Zhou, Irizarry and Li report the findings of a comprehensive review of the body of literature pertaining to the application of advanced technologies to construction health and safety spanning the period from 1986 to 2012. A total of 119 papers met the selection criteria for inclusion. The authors present a bibliographic analysis of this literature. They report the number of papers published annually, the publication type, the nature of the advanced technology deployed and the purpose for which it was deployed, the country/region of origin of the research, the research level, the phase of construction targeted by the technology and the project type. The authors note an increasing diversity in the type of advanced technology applied to construction safety and a shift from reactive to proactive usage of this technology. They also note there has been less emphasis on using technology to improve safety during the preconstruction and maintenance stages of a project. There has been little attempt to understand the cost effectiveness, unintended consequences and legal issues pertaining to the use of advanced technology for safety purposes. Finally, they identify a need to undertake more extensive research into factors influencing the adoption of advanced technology into industry practice and recommend that future researchers address this issue.

Sherratt, Farrell and Noble investigated how safety is socially constructed on large UK construction sites by those who work there. The authors sought an alternative perspective on safety through a discourse analysis of the text, talk and signage of safety, gathered from sites. They explore two of the emergent master discourses of safety in detail: safety as enforcement and safety as engagement. They suggest that these discourses reflect a shift in safety management over recent years, from a command-driven style to an approach embedded in engagement and personalization of safety for the workforce. The application of social constructionism revealed considerable variation in how people position safety within their daily work practice. Social constructionism also highlighted the continued influence of hierarchical management structures in terms of workforce behaviours and attitudes to safety, and an operational environment in which violations are the accepted norm. Such a context is incompatible with current safety management approaches, in terms of responsibility and ownership of safety. These findings are important to academics and practitioners and provide deeper insight into existing social constructions of safety, which can subsequently support the development of more compatible safety interventions through further research and practice.

Marks and Teizer challenge the definition and practice of real time proactive safety in the construction and material handling industries. Too much proximity between heavy construction equipment and pedestrian workers can lead to fatalities or severe injuries. The authors have developed a technology that issues warnings and alerts before an accident occurs. They demonstrate and validate a method for testing such a technology in construction operations. They present numerous field experiments which emulate typical interactions between pedestrian workers and construction equipment. Experimental results show that proximity detection and alert technologies can reliably provide alerts to equipment operators at different pre-calibrated proximity alert ranges. The authors argue that employing this technology has the potential to reduce injuries and fatalities and minimise collatoral damage associated with collisions between equipment and people. Their findings suggest changes in existing safety standards, management and engineering to effectively protect the construction workforce from hazards that arise as a result of the dynamic construction environment.

Forman examines the formation and stabilization of the coupling between the lean construction tool 'Last Planner System' and health and safety work on three construction sites in Denmark. She focuses on social processes rather than features of the technology. She examines how historical and contextual factors influence the formation and stabilization of the coupling between health and safety work and use of the Last Planner System. She found that the stabilized new practices were dependent on the relationship between functional departments at the company and the construction project, the organizational location of the key actors that drive the change programmes, the softness of the concepts that are implemented and the perception of health and safety. These findings challenge the assumption that features of lean construction can resolve onsite health and safety problems. These results can serve to guide future research to include change theories in the perception of the relation between lean construction and health and safety work on site. This work may also inform the design of health and safety and production planning strategies that facilitate onsite health and safety in construction projects.

Melzner, Zhang, Teizer and Bargstädt describe an automated rule-checking application for falls from height. Fall-related construction accidents continue to cause serious injuries and fatalities on construction sites around the world and OHS regulations typically specify measures for the elimination of fall hazards in construction. However, the authors contend that many design and construction planning teams have little understanding of local safety codes. In an increasingly globalized industry, they suggest that this presents a significant problem, especially given regional variation in OHS regulations and practices. The authors present a tool that provides the automatic detection of construction fall hazards inherent in a planned sequence of construction activities. The customizable, automated, safety, rule-checking system utilizes commercially available building information modelling (BIM) software. The system was tested and preliminary evidence suggests it was effective in the identification of fall hazards and the specification of solutions appropriate in two different regulatory contexts (Germany and the USA). The authors recommend that the role of BIM in augmenting the industry's safety design and planning processes requires further detailed investigation.

Larsen and Whyte present an understanding of how practitioners within the site team experience the safety challenges embedded within the design process. Their research is positioned against the rapidly changing landscape associated with digital design and information modelling and the potential role it can play in relation to site safety. Their case material is drawn from a project for the construction of a major UK railway station. This project offered rich data due to the extreme complexity of the project, comprising refurbishing and adaptation of listed buildings, new buildings, work on a live railway station and above a live underground railway station, all within the centre of London. The empirical data were gathered predominately through semi-structured interviews, initially with key stakeholders before a snowballing strategy was deployed as key players were identified and themes developed. In addition, safety documents and statistics were analysed in association with the emerging themes. Larsen and Whyte argue that, despite advances in digital design practices, the adverse connection between design and safe construction remains. This manifests itself through late design information, poor understanding of practical site processes by designers, and a persistently high level of late design changes.

Koch addresses safety culture in construction, which, if understood better, could lead to improved prevention of one of the worst problems of the industry, i.e. occupational accidents. Koch's contribution adds to the growing body of qualitative studies of safety culture. He used an ethnographic framework involving the integration, differentiation, ambiguity and multiple configuration elements of safety cultures. The multiple configuration in the analysis spans from cultures at a carpenter's crew to constellations of national safety cultures. At the carpenter's crew Koch finds an overarching common integrative culture of pride of work, which overlaps with the differentiation of four cultures, and finds ambiguous perceptions of risk and possibilities for prevention. Using studies of 25 different safety cultures, indications of a national pattern of reactive and proactive safety cultures are revealed. Koch argues that future prevention effort needs to resonate with this complex pattern of safety cultures.

Conclusion

In some way, all of these papers indicate that OHS should not be decoupled from an organization's 'main game', i.e. the management of a business or project. The papers in the special issue provide an insight into some of the organizational, technological and cultural challenges inherent in the management of OHS in construction projects. The body of research presented also suggests some potentially important new lines of inquiry. The multiplicity of research methods and approaches reflected in the special issue is also very promising. The construction OHS challenge is complex and multi-faceted, and as such requires a nuanced response. Decomposing and addressing a single aspect of the problem is unlikely to bring about significant improvements. Rather, the integration of multiple approaches is critical to delivering improvements in construction OHS in the 'fourth age' of safety.

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